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Investigating temporary capture in the Sun–Jupiter three-body system via Lagrangian coherent structures

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ABSTRACT

The temporary capture (TC) of Jupiter-family objects has long been a pivotal focus in celestial mechanics research. This study investigates the TC of objects near Jupiter within the context of the planar circular restricted three-body problem (PCRTBP), employing Lagrangian coherent structures (LCSs) and periapsis Poincaré maps. Initially, LCSs are identified via periapsis Poincaré maps and applied to segment the phase space. Parameter scanning enables a detailed analysis, classifying the orbital behaviours of objects in the proximity of Jupiter into three distinct categories: TC, low-energy flyby, and collision, each designating specific regions in phase space. Subsequently, a novel method for screening potential TC objects within the Jupiter system is proposed and validated, informed by the dynamic characteristics of TC motions. The efficacy of this method is illustrated by the re-identification of six known TC comets and the prediction of a prospective TC asteroid, 2002 GV28. Within the framework of the PCRTBP, analogous TC trajectories for these comets and asteroids are identified, offering novel insights into the dynamics of TC events.

Key words: methods: numerical – celestial mechanics – comets: general – minor planets, asteroids: general.

1 INTRODUCTION

Jupiter, with its intricate satellite system, has become a crucial research subject for exploring planetary formation and evolution processes (Sun et al. 2023). Moreover, as a crucial gateway to the outer Solar system, Jupiter has attracted the attention of numerous exploration missions (Yang et al. 2023). Historically, Pioneer 10 (Hall 1974) and 11 (Anderson et al. 2002) conducted flybys of Jupiter in 1973 and 1974, respectively, providing the first detailed data on Jupiter's atmosphere and magnetic field. Subsequently, Voyager 1 (Kohlhase & Penzo 1977) and 2 (Stone & Miner 1989) further explored Jupiter's magnetic field, ring system, and satellites in 1979. The Galileo spacecraft reached Jupiter in 1995, orbited the planet for eight years, and released a probe into Jupiter's atmosphere, achieving the first direct measurements of Jupiter's atmospheric layers (D'Amario, Bright & Wolf 1992). The New Horizons spacecraft also conducted scientific observations during its flyby in 2007, using Jupiter's gravity assist to accelerate towards Pluto (Guo & Farquhar 2008). The Juno spacecraft entered Jupiter's orbit in 2016, dedicated to studying the planet's internal structure, atmospheric composition, magnetic and gravitational fields, as well as its origin and evolutionary history (Grassi et al. 2021).

The Jupiter system is valued not only for its significance to exploration but also for its complex dynamical behaviours, which render it a focal point of research (Astakhov & Farrelly 2004; Marzari & Scholl 2007; Qi & de Ruiter 2021). For instance, the temporary capture (TC) of Jupiter family comets, such as 39P/Oterma (Duarte & Jorba 2022), P/1996 R2 (Hahn & Lagerkvist 1999), 82P/Gehrels 3 (Epifani, Palumbo & Colangeli 2009), 147P/Kushida–Muramatsu (Ohtsuka et al. 2008), 74P/Smirnova-Chernykh (Carusi et al. 1984), and 111P/Helin–Roman–Crockett (Howell, Marchand & Lo 2001), entails these comets entering from beyond the Jupiter system, executing multiple flybys of Jupiter, and ultimately escaping. Research on the trajectories of such TC comets can help achieve low-energy transfers, low-cost captures, and expanded exploration opportunities.

The circular restricted three-body problem (CRTBP) provides an effective model for simplifying the motion of objects under the gravitational influence of two celestial bodies, particularly suitable for studying the motion of objects near Jupiter. On this basis, libration points, periodic orbits near libration points, and invariant manifolds offer powerful approaches for researching temporary satellite captures in the Jupiter system. Heppenheimer (1975), Horedt (1976), and Heppenheimer & Porco (1977) discussed the possibility of satellite capture through the L_1 gateway in the CRTBP. Koon et al. (2000) found that the trajectory of the TC comet 39P/Oterma in the Sun–Jupiter rotating frame exhibits similar characteristics to the invariant manifolds associated with planar Lyapunov orbits, constructing a heteroclinic–homoclinic chain closely resembling the comet's path. Additionally, Howell et al. (2001) revealed the close association between the stable and unstable manifold arcs related to Halo orbits and the trajectories of comets 39P/Oterma and 111P/Helin–Roman–Crockett. Haapala & Howell (2013) utilized the periapsis Poincaré map to display structures related to invariant manifolds, further exploring the phenomenon of temporary satellite capture. Based on this, they constructed solutions with similar characteristics to the paths of comets P/1996 R2, 82P/Gehrels 3,

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and 147P/Kushida–Muramatsu, and built TC trajectories similar to that of comet 111P/Helin–Roman–Crockett using three-dimensional invariant manifolds in spatial CRTBP. Paskowitz & Scheeres (2006) also studied TC trajectories in the Hill's three-body problem using the periapsis Poincaré map. Davis & Howell (2012) conducted a detailed investigation on the escape and collision trajectories near the smaller primary in the restricted problem of three bodies, based on the periapsis Poincaré map.

Previous studies have shown a close connection between temporary satellite capture and invariant manifolds. While the study of invariant manifolds linked to libration point orbits has dominated much of the research in this field, invariant manifolds connected to other periodic orbit families have received comparatively little focus (Russell & Lam 2007). The considerable diversity among periodic orbit types poses significant challenges in identifying their corresponding global invariant manifolds. As an alternative, Lagrangian coherent structures (LCSs) may be regarded as the non-autonomous analogues of invariant manifolds (Qu, Lin & Xu 2020), obtained by identifying ridges composed of local maxima in finite-time Lyapunov exponent (FTLE) maps (Green, Rowley & Haller 2007; Haller 2015).

In recent years, FTLE maps and their corresponding LCSs have been widely applied in trajectory design and dynamical behaviour analysis. FTLE maps illustrate the separation between neighbouring trajectories, thereby identifying stretching directions in phase space, which is particularly beneficial for designing low-energy transfer trajectories (Short et al. 2015; Canales et al. 2022a; Muralidharan & Howell 2023). For example, in complex multibody systems, such as those formed by a planet and its moons, FTLE maps serve as powerful tools for designing transfer trajectories between moons. Short et al. (2015) utilized FTLE maps to design transfer trajectories from Oberon to Titania in the Uranian system. Canales, Howell & Fantino (2021b) proposed a Moon-to-Moon Analytical Transfer (MMAT) method, efficiently determining the feasibility of transferring between two moons orbiting near a common planet using simplified models and analytical constraints (Canales, Howell & Fantino 2020). They further developed the MMAT method by incorporating additional impulsive maneuvers, thereby extending its applicability (Canales, Howell & Fantino 2022b). Subsequently, Canales et al. (2023) combined FTLE maps with the MMAT method, enabling various trajectory modes at the transfer endpoints, such as captures, landings, transits, and takeoffs, greatly expanding the types of transfers between moons (Canales, Howell & Fantino 2021a). Besides, Short & Howell (2014) identified LCSs in multibody systems to determine the stable and unstable manifolds of libration point periodic orbits, designing transfer trajectories from LEO to L_1 periodic orbits in the Earth–Moon system based on invariant manifolds.

LCSs, as ordered features in FTLE maps, can serve as boundaries for different regions of dynamical behaviour. Kikuchi et al. (2019) utilized FTLE maps to analyse the stability of coupled orbit-attitude motion near asteroids and identified the stability boundaries of attitude motion within these maps. Qu et al. (2020) have applied LCSs to the dynamics analysis of the planar parabolic/hyperbolic restricted three-body problem, demonstrating how LCSs can be used to identify and study natural escape and capture trajectories in these systems. Shang, Wu & Cui (2017) investigated escape, capture, flyby, and collision regions in asynchronous binary asteroid systems based on the partitioning effect of LCSs on phase space. Tyler & Wittig (2022) proposed a novel DA-LCS method for constructing three-dimensional LCSs in the elliptic restricted three-body problem, facilitating the distinction of different behaviour regions without any prior knowledge. These studies show that LCSs are powerful tools for analysing the dynamics of multibody systems.

In past exploration missions of Jupiter, significant attention has been directed towards the periapsis state of the spacecraft relative to Jupiter (Brasil et al. 2015). This is due to the understanding that maneuvers at this point can significantly alter the trajectory of the spacecraft, with capture maneuvers typically being executed. Motivated by previous research, both LCS and periapsis Poincare maps are employed to examine the motion of objects near Jupiter in this paper. This technique replaces the typically used position–velocity maps, providing a better representation of the influence of the periapsis state on TC events. The planar circular restricted three-body problem (PCRTBP) is employed, and LCSs, identified on the periapsis Poincare maps, are utilized to precisely segment the periapsis phase space into various subsets. The types of object motion corresponding to these subsets are then thoroughly analysed, leading to the identification of the geometric structures of TC, low-energy flyby, and collision motions within the periapsis phase space. Then, a screening method is developed based on the characteristics of TC motion. By analysing the projection of TC motion in the heliocentric inertial frame, the range of orbital elements where TC events may occur is established. This allows for precise screening of Jupiter-family objects within a specific range, significantly reducing computational costs. The effectiveness of this screening method is validated by the identification of six known TC comets and the revelation of the potential TC of the outer main-belt asteroid 2002 GV28 by Jupiter. Additionally, the uncertainty in the initial state of 2002 GV28 is considered. Within the context of the PCRTBP, PCRTBP trajectories similar to the path characteristics of these TC objects are also discovered.

The structure of this paper is organized as follows: the primary dynamical models discussed in this study are introduced in Section 2. The method for generating LCSs under the Sun–Jupiter PCRTBP is described in Section 3. The partitioning of phase space by LCSs is analysed, and the types of motion corresponding to the subsets of the partitioned phase space are investigated in detail in Section 4. In Section 5, a screening method for Jupiter's TC objects is proposed. Finally, the conclusions of this study are presented in Section 6.

2 BACKGROUND

In this section, we introduce two dynamical models relevant to our study: the PCRTBP and the high-fidelity ephemeris (HFE) model.

2.1 The PCRTBP model

Most known objects that can be temporarily captured by Jupiter, such as 82P/Gehrels 3 and P/1996 R2, have orbital inclinations in the heliocentric J2000.0 ecliptic coordinate system of approximately 1.12° and 2.60° (Ohtsuka et al. 2008), respectively, which are very close to Jupiter's orbital inclination of about 1.30°. Consequently, the orbits of these objects can be approximated as coplanar with Jupiter's orbital plane. Based on this fact, the well-known PCRTBP (Poincaré 1967) is chosen as the primary model for this study, with detailed descriptions provided by Alessi, Gómez & Masdemont (2009). Additionally, selecting the planar scenario significantly reduces computational complexity, and as a preliminary analysis, this model remains highly effective in capturing the main physical phenomena and dynamical behaviours. This simplification allows us to gain a clearer understanding of the TC mechanism.

To ensure clarity, we present a brief overview of this model. Within this framework, the mass-negligible object P_3 moves within

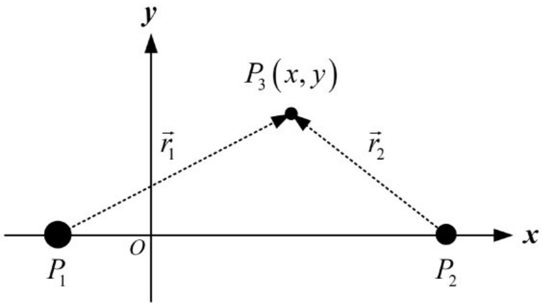


Figure 1. The sketch of the PCRTBP.

the gravitational field formed by the primary body P_1 and the secondary body P_2, where the motions of P_1 and P_2 are idealized as circular orbits around their common centre of mass. The motion of the mass-negligible object P_3 is confined to the plane of these circular orbits, as illustrated in Fig. 1. According to the technique of dimensionless units defined in Gomez & Olle (1991), the positions of P_1 and P_2 in the barycentric rotating (BCR) frame are set as $(-\mu, 0)$ and $(1-\mu, 0)$, respectively, where $\mu = m_2/(m_1+m_2)$, with m_1 representing the mass of P_1 and m_2 the mass of P_2. In this study, the Sun–Jupiter PCRTBP is used to analyse the TC dynamics within the Jupiter system. Therefore, P_1 represents the Sun, P_2 represents Jupiter, and the value of μ is set to 9.5388609×10^{-4}.

$$\begin{cases} \ddot{x} - 2\dot{y} = \partial\Omega/\partial x \\ \ddot{y} + 2\dot{x} = \partial\Omega/\partial y, \end{cases} \tag{1}$$

where the spacecraft's position vector is denoted by $\boldsymbol{r} = [x, y]^T$, and its velocity vector by $\boldsymbol{v} = [\dot{x}, \dot{y}]^T$. The superscript T indicates the transpose of a matrix, and Ω represents the system's pseudo-potential

$$\Omega = \frac{1}{2}\left(x^2+y^2\right) + \frac{1-\mu}{r_1} + \frac{\mu}{r_2} + \frac{1}{2}\mu(1-\mu), \tag{2}$$

where r_1 and r_2 represent the distances from the object to the Sun and from the object to Jupiter, respectively. These distances can be expressed as

$$\begin{cases} r_1 = \sqrt{(x+\mu)^2 + y^2} \\ r_2 = \sqrt{(x-1+\mu)^2 + y^2}. \end{cases} \tag{3}$$

In the PCRTBP model, the equations of motion include a unique energy integral, referred to as the Jacobi integral C, which is expressed as

$$C = 2\Omega - \dot{x}^2 - \dot{y}^2. \tag{4}$$

2.2 The high-fidelity ephemeris model

In the HFE model (Canales et al. 2021b), the equations of motion for the object are modelled in the heliocentric J2000.0 ecliptic coordinate system, considering multiple perturbing bodies with masses $\hat{m}_q$, where $q = 1, \ldots, n$.

$$\ddot{\boldsymbol{r}}_{ps} = \frac{-Gm_1}{r_{ps}^3}\boldsymbol{r}_{ps} + G\sum_{q=1}^{n}\hat{m}_q\left(\frac{\boldsymbol{r}_{sq}}{r_{sq}^3} - \frac{\boldsymbol{r}_{pq}}{r_{pq}^3}\right), \tag{5}$$

where the first term denotes the gravitational acceleration exerted by the Sun on the object P_3, while the summation term covers the gravitational interactions between the perturbing bodies, P_3, and the Sun.

G is the gravitational constant. The position vector from the Sun to P_3, within the heliocentric J2000.0 ecliptic coordinate system, is represented by $\boldsymbol{r}_{ps}$; the position vector from the Sun to the qth perturbing body is represented by $\boldsymbol{r}_{pq}$; and the position vector from P_3 to the qth perturbing body is represented by $\boldsymbol{r}_{sq}$. The formulation of the HFE model is based on the Jet Propulsion Laboratory's Development Ephemerides (DE440, Park et al. 2021), which provides precise positions and velocities for most bodies in the Solar system, enabling the derivation of $\boldsymbol{r}_{ps}$, $\boldsymbol{r}_{pq}$, and $\boldsymbol{r}_{sq}$ in equation (5).

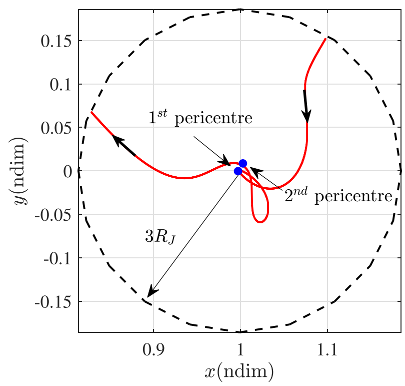


Figure 2. The ephemeris path of 82P/Gehrels 3 in the Sun–Jupiter rotating frame.

Then, we conduct a detailed analysis of the motion of comet 82P/Gehrels 3 near Jupiter, considering the gravitational influences of various celestial bodies, including the Sun, Venus, Mercury, Earth, Moon, Mars, Jupiter, Saturn, Uranus, and Neptune, as well as the gravitational effects of Jupiter's major moons, Io, Europa, Ganymede, and Callisto. The aim is to determine which bodies' gravitational forces predominantly affect the motion of comet 82P/Gehrels 3 near Jupiter. The ephemeris path of 82P/Gehrels 3 is obtained from the Jet Propulsion Laboratory (JPL) Horizons database (https://ssd.jpl.nasa.gov/horizons) and transformed into the Sun–Jupiter rotating frame (Haapala & Howell 2013). The trajectory is shown in Fig. 2, where the red curve represents the comet's path, and the blue dot indicates the pericentre with respect to Jupiter. According to the studies by Granvik, Vaubaillon & Jedicke (2012) and Qi & de Ruiter (2018), TC occurs within a sphere centred on the planet with a radius three times the planet's Hill radius. Therefore, in Fig. 2, we only display the trajectory of comet 82P/Gehrels 3 within a disc region centred on Jupiter, with a radius three times Jupiter's Hill radius, $R_H = 3R_{\mathrm{soi}}$. Here, R_{soi} denotes Jupiter's Hill radius. The trajectory of 82P/Gehrels 3 within this disc region contains two pericentres with respect to Jupiter, as shown in Fig. 2.

Fig. 3 shows the forces acting on comet 82P/Gehrels 3 while it is within this disc region. Specifically, the magnitude of the gravitational acceleration vectors exerted by different bodies (the Sun, other planets, and Jupiter's moons) on the comet is denoted by a_{body}, where the subscript 'body' indicates the specific perturbing body.

From Fig. 3, it is evident that the gravitational influences of the Sun and Jupiter dominate the TC motion of 82P/Gehrels 3. Near the first and second pericentre, the gravitational forces of Jupiter's moons increase significantly, reaching a magnitude comparable to that of the Sun's gravity at the first pericentre. Therefore, the gravitational influence of Jupiter's moons cannot be neglected. Additionally, Fig. 3 shows that the gravitational forces exerted by Venus, Mercury, Earth, and Saturn are relatively close, around 10^{-11} km s^{-2}. In contrast, the gravitational forces of Mars, the Moon, Uranus, and

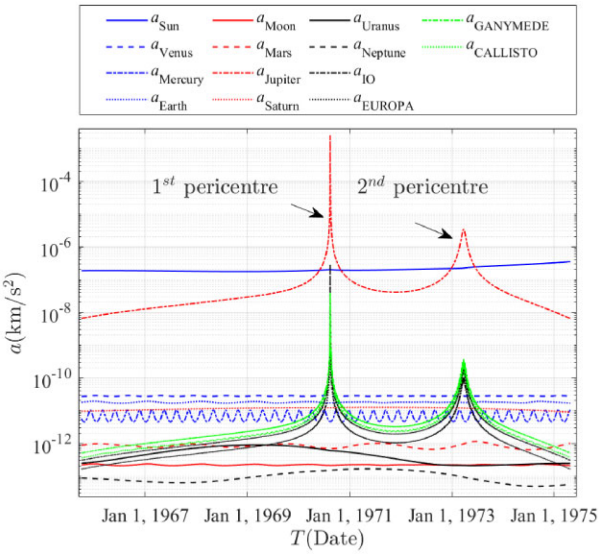


Figure 3. The forces acting on 82P/Gehrels 3 during TC by Jupiter.

Neptune fluctuate between 10^{-12} and 10^{-13} km s^{-2}, which are one to two orders of magnitude smaller than those of Earth and Saturn. Consequently, their influence on the motion of 82P/Gehrels 3 is negligible. Based on the above analysis, the ephemeris model in this study considers the gravitational influences of the Sun, Jupiter, Saturn, Earth, Venus, Mercury, and the four major moons of Jupiter: Io, Europa, Ganymede, and Callisto.

3 LAGRANGIAN COHERENT STRUCTURES

This section introduces the FTLE, presents the periapsis Poincaré map for analysing the motion of objects near Jupiter, and describes the method for calculating the LCSs within this map.

3.1 The finite-time Lyapunov exponent

In the realm of multibody dynamics, the FTLE quantifies the divergence rate of neighbouring trajectories within a dynamical system. This exponent is invaluable for elucidating the orbital motion characteristics of an object located near Jupiter, especially when considering the Sun's gravitational influence. For the dynamical system outlined in equation (1), given the object's initial position at time t_0 as $\boldsymbol{r}(t_0) = [x(t_0), y(t_0)]^T$, its position at a later time $t_0 + \Delta T$ is denoted by $\phi_{t_0}^{\Delta T}(\boldsymbol{r}(t_0))$, where ϕ denotes the manifold mapping of equation (1). Supposing an infinitesimal perturbation $\delta \boldsymbol{r}(t_0)$ is applied to $\boldsymbol{r}$ at time t_0, the corresponding variation at time $t_0 + \Delta T$ is given by (Qu et al. 2020)

$$\begin{aligned}\delta \boldsymbol{r}(\Delta T) &= \phi_{t_0}^{\Delta T}(\boldsymbol{r}(t_0) + \delta \boldsymbol{r}(t_0)) - \phi_{t_0}^{\Delta T}(\boldsymbol{r}(t_0)) \\ &= \frac{\partial \phi_{t_0}^{\Delta T}(\boldsymbol{r}(t_0))}{\partial \boldsymbol{r}(t_0)} \cdot \delta \boldsymbol{r}(t_0) + O\left(\delta \boldsymbol{r}^2(t_0)\right). \end{aligned} \quad (6)$$

By disregarding the higher order infinitesimal terms, the resulting equation is as follows

$$\|\delta \boldsymbol{r}(\Delta T)\| = \sqrt{\langle \delta \boldsymbol{r}(t_0), \Delta \cdot \delta \boldsymbol{r}(t_0)\rangle}, \quad (7)$$

where Δ denotes the Cauchy–Green deformation tensor, which is defined as follows

$$\Delta = \left(\frac{\partial \phi_{t_0}^{\Delta T}(\boldsymbol{r}(t_0))}{\partial \boldsymbol{r}(t_0)}\right)^T \cdot \left(\frac{\partial \phi_{t_0}^{\Delta T}(\boldsymbol{r}(t_0))}{\partial \boldsymbol{r}(t_0)}\right), \quad (8)$$

where $\partial \phi_{t_0}^{\Delta T}(\boldsymbol{r}(t_0))/\partial \boldsymbol{r}(t_0)$ represents the spatial gradient tensor of the flow $\phi_{t_0}^{\Delta T}(\boldsymbol{r}(t_0))$. Further, it can be obtained that (Qu et al. 2020)

$$\max_{\delta \boldsymbol{r}(t_0)} \|\delta \boldsymbol{r}(\Delta T)\| = \sqrt{\lambda_{\max}} \cdot \boldsymbol{\xi} \quad (9)$$

were $\lambda_{\max}$ represents the largest eigenvalue of Δ, with $\boldsymbol{\xi}$ being its associated eigenvector. Consequently, the FTLE is defined as follows

$$\sigma_{t_0}^{\Delta T}(\boldsymbol{r}(t_0)) = \frac{1}{|\Delta T|} \ln \sqrt{\lambda_{\max}(\Delta)}. \quad (10)$$

If $\sigma_{t_0}^{\Delta T}(\boldsymbol{r}(t_0)) \leq 0$, it indicates that adjacent objects converge to a single point, corresponding to an equilibrium point or periodic motion. Conversely, if $\sigma_{t_0}^{\Delta T}(\boldsymbol{r}(t_0)) > 0$, it signifies that adjacent objects will eventually diverge, corresponding to local instability in the trajectories. Using equations (8) and (10), we can calculate the FTLE values for each object arranged near Jupiter at time t_0, thereby obtaining the FTLE map.

The FTLE can be obtained through either forward or backward integration. If $\Delta T > 0$, this is referred to as a forward FTLE map; the larger the $\sigma_{t_0}^{\Delta T}(\boldsymbol{r}(t_0))$, the greater the repulsion between adjacent trajectories. Conversely, if $\Delta T < 0$, it is called a backward FTLE map; the larger the $\left|\sigma_{t_0}^{\Delta T}(\boldsymbol{r}(t_0))\right|$, the greater the attraction between adjacent trajectories. By setting a certain threshold of $\sigma_{t_0}^{\Delta T}(\boldsymbol{r}(t_0))$, ridges composed of local maxima in the FTLE map can be extracted, which are commonly referred to as LCSs. In fluid dynamics, LCSs serve as boundaries separating regions with fundamentally different behaviours. For $\Delta T > 0$, LCSs can be considered as non-autonomous analogues of stable manifolds, while for $\Delta T < 0$, LCSs can be considered as non-autonomous analogues of unstable manifolds (Qu et al. 2020). In this study, when objects are initially located near Jupiter, LCSs may effectively delineate regions characterized by distinct dynamical behaviours.

3.2 Periapsis Poincaré maps

The periapsis Poincaré map is a technique used to capture the dynamical behaviour of an orbiting object as it passes through periapsis. The periapsis is the point in a multibody system where the object is closest to the primary body, at which the orbit reaches its minimum radial distance relative to the primary, followed by a change in the flight path angle. By using Poincaré sections, this map projects the system's state onto a lower dimensional space (typically two or three dimensions), thereby simplifying the complex high-dimensional orbital dynamics and highlighting the crucial states at periapsis. This approach helps to more clearly reveal the periodicity, stability, and chaotic characteristics of the object's motion. It is important to note that the periapsis Poincaré section must satisfy the following conditions: the object's radial velocity relative to the planet must be zero, and the radial acceleration must be positive, as shown in Fig. 4. Given the advantages of this technique in simplifying and revealing the underlying dynamics, periapsis Poincaré maps are commonly employed to investigate the escape and capture dynamics of objects in proximity to the smaller primary within the CRTBP model. For example, Villac & Scheeres (2003) and Paskowitz & Scheeres (2006) utilized these maps to analyse the short-term behaviour of escape and capture trajectories in the context of Hill's three-body dynamics, specifically focusing on the Jupiter–Europa system. These studies emphasized the importance of understanding these behaviours, as they play a crucial role in designing low-energy transfer strategies for missions involving planetary moons.

To characterize the periapsis state of the object, we establish two coordinate systems: a polar coordinate system and a Jupiter-centred

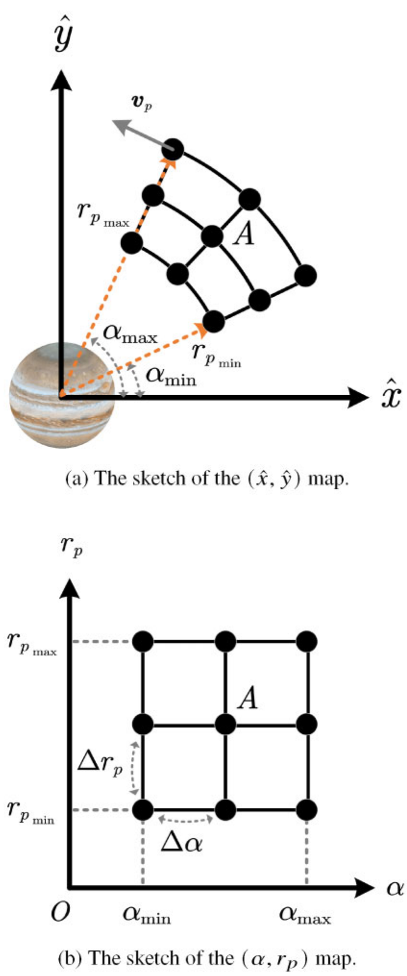


Figure 4. Sketches of periapsis Poincaré maps.

rotating (JCR) frame. This section aims to define the necessary parameters and transformations crucial for analysing the object's motion near Jupiter.

In the polar coordinate system, illustrated in Fig. 4(a), the periapsis state is characterized by two parameters: the periapsis distance r_p between the object and Jupiter, and the polar angle α. The angle α is defined as the angle between the vector $\boldsymbol{r}_p$, extending from Jupiter to the object, and the $\boldsymbol{x}$-axis of the Sun–Jupiter BCR frame. The counterclockwise rotation around Jupiter indicates direct motion ($\dot{\alpha} > 0$), while clockwise rotation indicates retrograde motion ($\dot{\alpha} < 0$).

For the JCR frame, shown in Fig. 4(a), the axes of this frame align with those of the Sun–Jupiter BCR frame. In this frame, the position and velocity vectors of the object are denoted as $\hat{\boldsymbol{r}}(t_0)$ and $\hat{\boldsymbol{v}}(t_0)$, respectively.

Notably, $\hat{\boldsymbol{r}}(t_0)$ and $\hat{\boldsymbol{v}}(t_0)$ also represent the position and velocity vectors of the object relative to Jupiter. The transformation between these vectors and the position $\boldsymbol{r}(t_0)$ and velocity $\boldsymbol{v}(t_0)$ vectors of the object relative to the Sun–Jupiter barycentre in the Sun–Jupiter BCR frame is given as follows

$$\begin{cases} \boldsymbol{r}(t_0) = \hat{\boldsymbol{r}}(t_0) + [1-\mu, 0]^T \\ \boldsymbol{v}(t_0) = \hat{\boldsymbol{v}}(t_0). \end{cases} \tag{11}$$

Subsequently, the transformation matrix between the JCR frame and the polar coordinate system is introduced. First, the position vector $\hat{\boldsymbol{r}}(t_0)$ of the object in the JCR frame can be described by the parameters r_p and α in the polar coordinate system, which is independent of the type of motion of the object, whether it is direct or retrograde. The position vector at time t_0 can be expressed as

$$\hat{\boldsymbol{r}}(t_0) = \begin{bmatrix} \hat{x}(t_0) \\ \hat{y}(t_0) \end{bmatrix} = \begin{bmatrix} r_p(t_0)\cos\alpha(t_0) \\ r_p(t_0)\sin\alpha(t_0). \end{bmatrix} \tag{12}$$

Then, the velocity transformation matrix between the two coordinate systems is introduced. For a given Jacobi constant C_0, the magnitude of the velocity vector $\hat{\boldsymbol{v}}(t_0)$ can be derived from the following equation

$$\hat{v}(t_0) = \sqrt{2\Omega(t_0)^2 - C_0}. \tag{13}$$

At periapsis, the position vector $\hat{\boldsymbol{r}}(t_0)$ and the velocity vector $\hat{\boldsymbol{v}}(t_0)$ are orthogonal. The velocity vector can be defined using the flight path angle $\beta(t_0)$. For direct motion ($\dot{\alpha}(t_0) > 0$), the flight path angle is given by $\beta(t_0) = \alpha(t_0) + \pi/2$; conversely, for retrograde motion ($\dot{\alpha}(t_0) < 0$), it is defined as $\beta(t_0) = \alpha(t_0) - \pi/2$. Thus, the velocity vector $\hat{\boldsymbol{v}}(t_0)$ can be expressed as

$$\hat{\boldsymbol{v}}(t_0) = \begin{bmatrix} \dot{\hat{x}}(t_0) \\ \dot{\hat{y}}(t_0) \end{bmatrix} = \begin{bmatrix} \hat{v}(t_0)\cos\beta(t_0) \\ \hat{v}(t_0)\sin\beta(t_0). \end{bmatrix} \tag{14}$$

In summary, using equation (11), we can achieve the state transformation of an object between the JCR and the BCR frames. Similarly, equations (12)–(14) allow us to perform the state transformation between the JCR frame and the polar coordinate system. Consequently, by combining equations (11)–(14), the state transformation between the polar coordinate system and the BCR frame can also be realized.

According to the aforementioned assumptions, in the Sun–Jupiter PCRTBP model, once the type of motion (whether direct or retrograde) is selected, the periapsis state in the polar coordinate system can be uniquely determined by r_p, α, and C_0. Additionally, in the JCR frame, the state space of the object is four-dimensional. When projected onto the periapsis section (where the radial velocity relative to Jupiter is zero and the radial acceleration is positive), the dimensionality of the state space is reduced by one. Consequently, the periapsis state can be uniquely described by the position components $\hat{x}$ and $\hat{y}$ along with the Jacobi constant C_0.

Thus, we can use a three-dimensional phase space, either (r_p, α, C_0) or $(\hat{x}, \hat{y}, C_0)$, to visualize the periapsis states of a large number of objects relative to Jupiter. When the Jacobi constant C_0 of the Sun–Jupiter system is specified, the dimensionality is reduced by one. Consequently, the periapsis states of the objects can be depicted using two-dimensional maps, referred to as (α, r_p) and $(\hat{x}, \hat{y})$ maps. These two-dimensional maps are also employed in this paper to visually present the distribution of FTLE.

3.3 Computing LCSs in FTLE maps

Based on the above discussion, the periapsis Poincaré map can be visualized separately by the (α, r_p) and the $(\hat{x}, \hat{y})$ maps. The following introduces the Algorithm 1 for generating LCSs in these two maps.

The first step is to determine the range of interest for generating the FTLE in the periapsis Poincaré map, specifically the size of the interval (α, r_p), and then discretize it. We consider the motion of an object within the region near Jupiter where $r_{p_{\min}} \le r_p \le r_{p_{\max}}$ and $\alpha_{\min} \le \alpha \le \alpha_{\max}$ at initial time t_0. This region is depicted as a sector in Fig. 4(a) and as a rectangle in Fig. 4(b). Then, we discretize r_p and α within their respective ranges into M and N points using step sizes Δr_p and $\Delta\alpha$, respectively, resulting in an $M \times N$ main grid. Figs 4(a) and (b) illustrate an example where $M = N = 3$.

Algorithm 1 Compute FTLE and LCSs

Input: $r_{p_{\min}}, r_{p_{\max}}, \alpha_{\min}, \alpha_{\max}, C_0, \Delta T$
Input: Grid sizes M, N and step sizes Δr_p, $\Delta\alpha$
Output: FTLE distribution on maps (α, r_p) and $(\hat{x}, \hat{y})$

1: **Initialize the Main Grid:**
2: **for** each $i = 1$ to M **do**
3: **for** each $j = 1$ to N **do**
4: $r_{p_{i,j}} = r_{p_{\min}} + (i-1) \times \Delta r_p$
5: $\alpha_{i,j} = \alpha_{\min} + (j-1) \times \Delta\alpha$
6: **end for**
7: **end for**
8: **Map Main Grid to JCR Frame:**
9: **for** each main grid point (i, j) **do**
10: Determine type of motion (direct or retrograde)
11: Compute mapped position $(\hat{x}_{i,j}, \hat{y}_{i,j})$ using C_0 and Eqs. 12-14
12: **end for**
13: **Transform JCR to BCR Frame:**
14: **for** each mapped position $(\hat{x}_{i,j}, \hat{y}_{i,j})$ **do**
15: Compute mapped state $\boldsymbol{x}_{i,j} = [\boldsymbol{r}_{i,j}, \boldsymbol{v}_{i,j}]$ using Eq. 11
16: **end for**
17: **Detect Grid Points Representing Collision Trajectories of Jupiter:**
18: **for** each mapped position $\boldsymbol{x}_{i,j}$ **do**
19: Propagate $\boldsymbol{x}_{i,j}$ for ΔT
20: Record distance l to Jupiter
21: **if** $l <$ Jupiter's radius **then**
22: Mark point A as a collision point
23: **end if**
24: **end for**
25: **Compute FTLE for Non-Collision Points:**
26: **for** each non-collision point $\boldsymbol{x}_{i,j}$ **do**
27: Compute derivatives $a_{11}, a_{12}, a_{21}, a_{22}$
28: Calculate $\Delta_{i,j}$
29: Determine FTLE $\sigma_{t_0}^{\Delta T}(\boldsymbol{r}_{i,j})$
30: **end for**
31: **Map FTLE Results:**
32: **for** each main grid point (i, j) **do**
33: Detect LCS using LCS generation tool
34: Label LCSs for $\Delta T > 0$ as LCS$_+$ and for $\Delta T < 0$ as LCS$_-$
35: Map FTLE and LCSs onto (α, r_p) map and $(\hat{x}, \hat{y})$ map
36: **end for**
37: **return** FTLE and LCSs distribution

For clarity, the input parameters required for Algorithm 1 are summarized as follows: parameters $r_{p_{\min}}$, $r_{p_{\max}}$, $\alpha_{\min}$, and $\alpha_{\max}$ represent the size of the (α, r_p) map; the Jacobian constant C_0 of the Sun–Jupiter system; the integration time duration ΔT necessary for calculating the FTLE, where $\Delta T > 0$ indicates forward-time propagation, and $\Delta T < 0$ indicates backward-time propagation; parameters M and N characterizing the grid size in the discretized (α, r_p) map; and the discretization steps Δr_p and $\Delta\alpha$.

In the discretized (α, r_p) map shown in Fig. 4(b), the coordinates of the gridpoint in the ith row (counting from left to right) and jth column (counting from bottom to top) are denoted as $(\alpha_{i,j}, r_{p_{i,j}})$. Here, i and j are indices of main gridpoints, which are positive integers, with $1 \le i \le M$ and $1 \le j \le M$. The specific values of $\alpha_{i,j}$ and $r_{p_{i,j}}$ are calculated using the formulae in lines 4 and 5 of Algorithm 1.

To calculate the FTLE values corresponding to the main gridpoints under the PCRTBP model, we need to obtain the initial values of the orbital propagation at time t_0. This requires mapping the discretized gridpoints from the (α, r_p) map in polar coordinates to the Sun–Jupiter BCR frame, with the JCR frame serving as an intermediary to facilitate this mapping. Below, we use the main gridpoint A (with coordinates $(\alpha_{i,j}, r_{p_{i,j}})$) in the (α, r_p) map as an example to illustrate this coordinate transformation process. After selecting the type of motion as either direct or retrograde, for a given Jacobi constant C_0, we can obtain the mapped position $(\hat{x}_{i,j}, \hat{y}_{i,j})$ of $(\alpha_{i,j}, r_{p_{i,j}})$ in the JCR frame using equations (12)–(14), as shown in Fig. 4(a). Subsequently, the object's periapsis state $(\hat{x}_{i,j}, \hat{y}_{i,j})$ in the JCR frame is transformed to the Sun–Jupiter BCR frame for FTLE computation using equation (11). The mapped position $(x_{i,j}, y_{i,j})$ of point A in the BCR frame is depicted in Fig. 5, with the mapped state represented as $\boldsymbol{x}_{i,j} = [\boldsymbol{r}_{i,j}, \boldsymbol{v}_{i,j}]$.

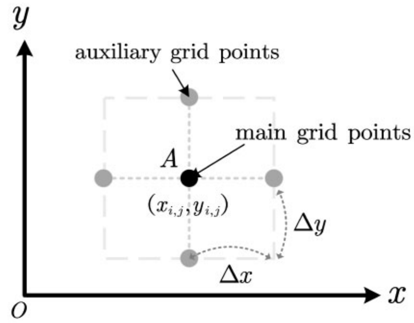


Figure 5. The sketch of the auxiliary grid.

Similarly, the calculation process for determining the FTLE value at the main gridpoint is demonstrated using point A as an example. Before computing the FTLE corresponding to point A, it is necessary to determine whether $\boldsymbol{x}_{i,j}$ corresponds to a collision trajectory with Jupiter. Consequently, $\boldsymbol{x}_{i,j}$ is propagated over a time interval ΔT, and the distance l from the object to Jupiter is recorded at each instant. If l is less than Jupiter's radius, $\boldsymbol{x}_{i,j}$ is deemed to correspond to a collision trajectory with Jupiter. In this study, if point A is designated as a collision point, then its FTLE is not calculated.

If point A is not identified as a collision point, the next step is performed to compute its FTLE. In current research on the FTLE, it is common practice to discretize $\partial\phi_{t_0}^{\Delta T}(\boldsymbol{r})/\partial\boldsymbol{r}$ in equation (8) using the finite-difference method

$$\frac{\partial\phi_{t_0}^{\Delta T}\left(\boldsymbol{r}_{i,j}\right)}{\partial\boldsymbol{r}_{i,j}} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}, \tag{15}$$

where

$$a_{11} = \frac{x_{i,j}\left(t_0 + \Delta T; x_{i,j} + \Delta x\right) - x_{i,j}\left(t_0 + \Delta T; x_{i,j} - \Delta x\right)}{2\Delta x}, \tag{16}$$

$$a_{12} = \frac{x_{i,j}\left(t_0 + \Delta T; y_{i,j} + \Delta y\right) - x_{i,j}\left(t_0 + \Delta T; y_{i,j} - \Delta y\right)}{2\Delta y}, \tag{17}$$

$$a_{21} = \frac{y_{i,j}\left(t_0 + \Delta T; x_{i,j} + \Delta x\right) - y_{i,j}\left(t_0 + \Delta T; x_{i,j} - \Delta x\right)}{2\Delta x}, \tag{18}$$

$$a_{22} = \frac{y_{i,j}\left(t_0 + \Delta T; y_{i,j} + \Delta y\right) - y_{i,j}\left(t_0 + \Delta T; y_{i,j} - \Delta y\right)}{2\Delta y}. \tag{19}$$

In this paper, the auxiliary grid technique has been employed, which locally refines the $M \times N$ main gridpoints obtained in the previous section, as illustrated in Fig. 5. Δx and Δy represent the grid spacings of the auxiliary grid along the $\boldsymbol{x}$- and $\boldsymbol{y}$-axes, respectively. It has been demonstrated by Farazmand & Haller (2012) that, compared to the FTLE fields computed using only the main grid, the auxiliary grid technique enhances the accuracy of equation (15) with only a marginal increase in computational effort, thereby improving the quality of the FTLE maps. It is suggested by experience that the spacing of the auxiliary grid should be set to 1 per cent–10 per cent of the main grid spacing (Onu, Huhn & Haller 2015). In this study, it is assumed that $\Delta x = \Delta y = \Delta r_p/100$. By propagating the initial states at time t_0 corresponding to auxiliary gridpoints under the PCRTBP model for a duration ΔT, a_{11}, a_{12}, a_{21}, and a_{22} are obtained. From equation (15), the discrete forms of equations (10) and (8) can be derived.

$$\Delta_{i,j} = \left(\frac{\partial \phi_{t_0}^{\Delta T}\left(\boldsymbol{r}_{i,j}\right)}{\partial \boldsymbol{r}_{i,j}}\right)^T \cdot \left(\frac{\partial \phi_{t_0}^{\Delta T}\left(\boldsymbol{r}_{i,j}\right)}{\partial \boldsymbol{r}_{i,j}}\right), \tag{20}$$

$$\sigma_{t_0}^{\Delta T}\left(\boldsymbol{r}_{i,j}\right) = \frac{1}{|\Delta T|} \ln \sqrt{\lambda_{\max}\left(\Delta_{i,j}\right)}. \tag{21}$$

In the Sun–Jupiter BCR coordinate system, for all non-collision points $\boldsymbol{x}_{i,j}$ within the main grid, the corresponding FTLE values are calculated according to the steps outlined in lines 27–29 of Algorithm 1. This yields the FTLE distribution in the periapsis Poincaré map under the Sun–Jupiter BCR frame. Similarly, using point A as an example, with its coordinates in the Sun–Jupiter BCR frame given as $\boldsymbol{x}_{i,j}$, the corresponding FTLE value is denoted as $f_{i,j}$. Through the inverse mapping of equation (11), the coordinates of this point in the JCR frame, denoted as $(\hat{x}_{i,j}, \hat{y}_{i,j})$, can be calculated, assuming the FTLE value remains $f_{i,j}$. By applying the inverse mappings from equations (12)–(14), the polar coordinates $(\alpha_{i,j}, r_{p_{i,j}})$ corresponding to point A can be determined, with the FTLE value remaining $f_{i,j}$. Subsequently, for each non-collision main gridpoint in the Sun–Jupiter BCR frame, the coordinates and corresponding FTLE values are mapped to the $(\hat{x}, \hat{y})$ map in the JCR frame and the (α, r_p) map in the polar coordinate. Similarly, the LCSs are first detected in the periapsis Poincaré map using the LCS generation tool developed by Onu et al. (2015) within the Sun–Jupiter BCR frame. The coordinates of the main gridpoints that constitute the LCSs are then mapped to the $(\hat{x}, \hat{y})$ and (α, r_p) maps through inverse mapping, resulting in the LCS distribution in both$(\hat{x}, \hat{y})$ and (α, r_p) maps. Here, LCSs detected in the FTLE maps for $\Delta T > 0$ and $\Delta T < 0$ are labelled as LCS_+ and LCS_-, respectively.

4 ANALYSING THE MOTION OF OBJECTS NEAR JUPITER USING LAGRANGIAN COHERENT STRUCTURES

This section begins with an analysis of the LCSs within the Sun–Jupiter PCRTBP model, which partition the phase space into several regions. A detailed investigation of the geometric characteristics of the trajectories represented by each region is conducted, categorizing the motion of objects near Jupiter into TC, low-energy flyby, and collision. Finally, the influence of the periapsis parameters and maneuvers of the object on the type of motion is examined.

4.1 Lagrangian coherent structures in the Sun–Jupiter PCRTBP

To generate FTLE maps under the Sun–Jupiter PCRTBP frame, the initial step involves defining the size of the (r_p, α) map. In this study, α ranges from 0° to 360°, with r_p extending from $4R_J$ to

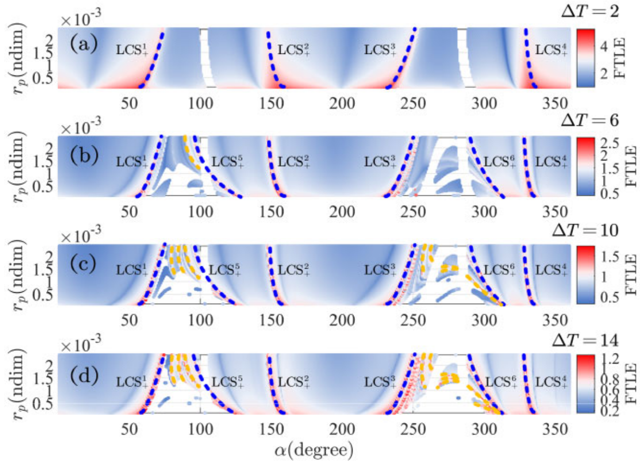


Figure 6. The FTLE maps with different ΔT.

the semimajor axis of Callisto's orbit, approximately 1 882 700 km. Then r_p is divided into 100 equidistant points, and α into 3600 equally spaced points. Observations from the Sun–Jupiter BCR frame indicate that comets temporarily captured by Jupiter, such as 147P/Kushida–Muramatsu, 82P/Gehrels 3, 111P/Helin–Roman–Crockett, and P/1996 R2, meet the condition $\dot{\alpha} > 0$ at their final pericentre with respect to Jupiter before escaping, suggesting escape on direct orbits (Ohtsuka et al. 2008). Therefore, in this research, we posit that the initial motion of objects at the pericentre is direct, resulting in $\beta = \alpha + \pi/2$.

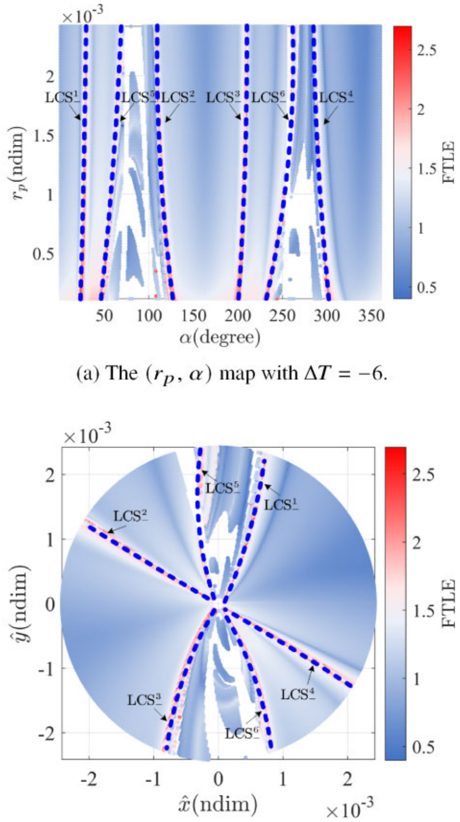
4.1.1 *Analysis of forward and backward time LCSs under a fixed Jacobi constant*

Based on the assumptions and the LCS calculation method outlined in Section 3.3, we propagate the discrete periapsis state of the object in the Sun–Jupiter BCR frame using the PCRTBP model. This allows us to compute FTLE values at various time instances, generating FTLE maps for different normalized ΔT values of 2, 6, 10, and 14, with C set to 3.03. Here, $\Delta T > 0$ indicates that the FTLE maps are obtained by propagating the periapsis state forward in time.

In Fig. 6, the blank areas represent the propagation trajectories of objects over time ΔT that collide with Jupiter. For $\Delta T = 2$, as shown in Fig. 6(a), four ridges appear in the FTLE map, labelled LCS_+^1 to LCS_+^4. When ΔT increases to 6, additional LCS, labelled LCS_+^5 and LCS_+^6, are identified in Fig. 6(b). Notably, LCS_+^5 is discontinuous due to truncation by the collision zone, but it is considered approximately continuous for simplicity since the collision zone occupies a small portion of it. As ΔT further increases to 10 or 14, more LCSs, indicated by non-continuous dashed lines, appear in Figs 6(c) and (d). These LCS are highly sensitive to ΔT, leading to the conclusion that we focus on those that remain stable across different ΔT values, represented by the blue dashed lines.

Comparing Figs 6(a) and (d), it is clear that a ΔT of 2 does not adequately capture the motion characteristics near Jupiter, as evidenced by the absence of LCS_+^5 and LCS_+^6 in Fig. 6(a). However, for ΔT values of 6, 10, or 14, the identified LCSs show consistent characteristics, allowing us to set ΔT to 6 for computational efficiency, with the backward propagation set to −6.

Subsequently, we obtained backward-propagated FTLE maps for $C = 3.03$, as illustrated in Fig. 7. Figs 7(a) and (b) display the



(a) The (r_p, α) map with $\Delta T = -6$.

(b) The $(\hat{x}, \hat{y})$ map with $\Delta T = -6$.

Figure 7. Periaspsis Poincaré maps with $\Delta T = -6$.

(α, r_p) and $(\hat{x}, \hat{y})$ maps, respectively. In these maps, six distinct ridges, labelled LCS_- from LCS_-^1 to LCS_-^6, are visible. Similar to the forward propagation results, LCS_-^5 is positioned between LCS_-^1 and LCS_-^2, while LCS_-^6 is located between LCS_-^3 and LCS_-^4.

4.1.2 Analysis of forward and backward time LCSs under a varying Jacobi constant

Simulation results indicate that for Jacobi constant values (C_0) below 3.00, distinguishing between LCS_+^1, LCS_+^5, and LCS_+^2 is challenging due to their close proximity. Conversely, for C_0 values above 3.03, the likelihood of collisions with Jupiter increases. To effectively capture the global structure of the LCS, we have chosen a range for C_0 from 3.00 to 3.03, with a step size of 0.001.

For each value of C_0, we compute the FTLE distribution in the (α, r_p) map using the method described in Section 3.3 under the PCRTBP model. Local maxima in the FTLE map, which correspond to the LCSs, are identified, and their coordinates (α, r_p, C_0) are recorded. We consider two scenarios for computing the LCSs: forward integration over time ($\Delta T = 6$) and backward integration over time ($\Delta T = -6$).

For forward-time integration with $\Delta T = 6$, six LCSs are obtained, labelled LCS_+^1 to LCS_+^6. When plotted together in the phase space (α, r_p, C_0), these form six distinct surfaces, as shown in Fig. 8(a). Since these surfaces do not intersect, we utilize MATLAB's Neural Network Toolbox to construct a simple feedforward neural network, which fits the scattered points into smooth surfaces, illustrated in Fig. 8(b).

Similarly, for backward-time integration with $\Delta T = -6$, another six LCSs are obtained, labelled LCS_-^1 to LCS_-^6. When these are plotted in the phase space (α, r_p, C_0), they also form six distinct surfaces, as shown in Fig. 8(c). Again, since these surfaces do not intersect, we apply the same neural network approach to fit these points into smooth surfaces, as depicted in Fig. 8(d).

We observe that for any C_0, LCS_+^5 is always located between LCS_+^1 and LCS_+^2, and LCS_+^6 is positioned between LCS_+^3 and LCS_+^4. Similarly, LCS_-^5 is situated between LCS_-^1 and LCS_-^2, while LCS_-^6 is located between LCS_-^3 and LCS_-^4.

4.1.3 Partitioning the phase space using LCSs

In the preceding analysis, we determined the maximum and minimum values for the Jacobi constant C_0, the polar angle α, and the periapsis distance r_p. These values define the phase space (α, r_p, C_0), which represents the initial periapsis states of objects, depicted as a rectangular parallelepiped in Fig. 9. The LCS surfaces (specifically LCS_+^1 to LCS_+^6 and LCS_-^1 to LCS_-^6, shown in Fig. 8) partition this phase space into 17 distinct regions.

The geometric characteristics of these regions are illustrated in Fig. 10. In this figure, R_1 represents the envelope formed by the phase space (α, r_p, C_0) and LCS_-^1. It is approximately symmetric to the envelope R_{17}, formed by the phase space and LCS_+^4, with respect to the plane at $\alpha = 180°$. Similar symmetric relationships are observed between pairs: R_2 and R_{16}, R_3 and R_{15}, R_4 and R_{12}, R_5 and R_{13}, R_6 and R_{14}, R_7 and R_{11}, and R_8 and R_{10}. Additionally, as shown in Fig. 10(i), R_9 is approximately symmetric about the plane $\alpha = 180°$.

4.2 Classification of object motions near the Jupiter based on LCSs

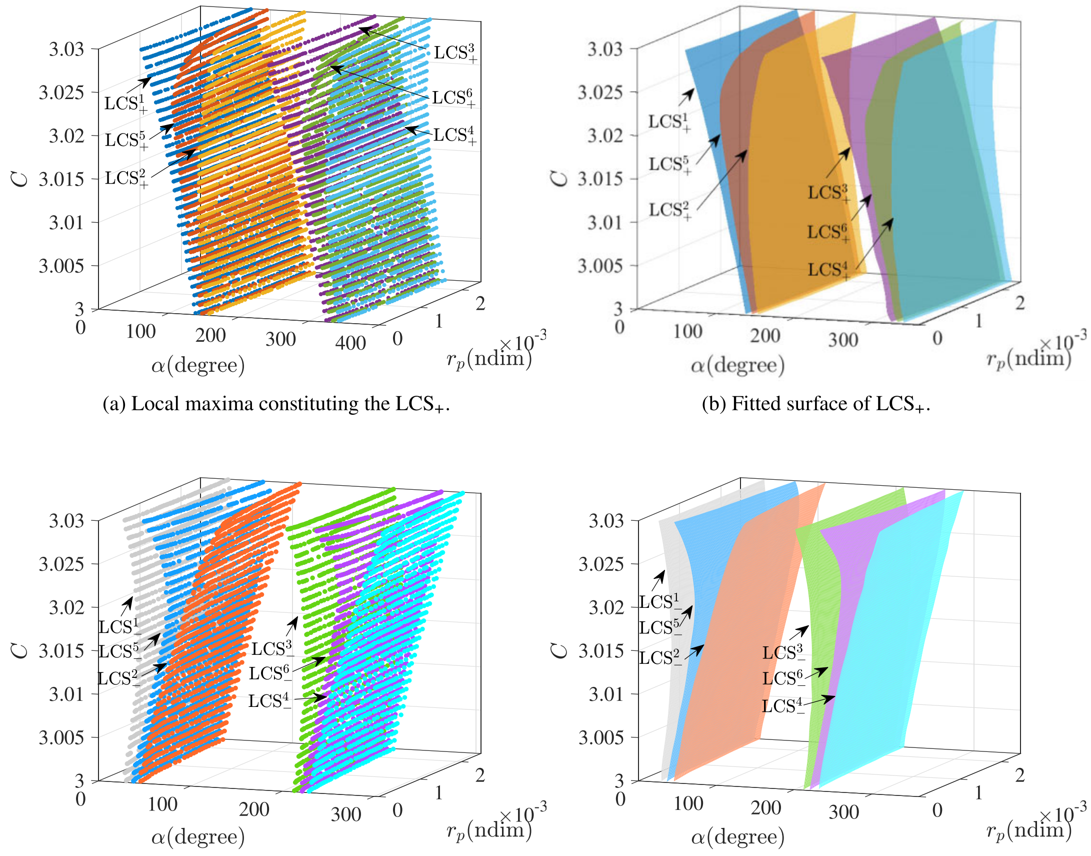
4.2.1 Types of motion of objects near Jupiter

The complex gravitational environment in the Sun–Jupiter system leads to various motion behaviours for objects near Jupiter. Based on previous studies on the motions of objects near planets (Howell et al. 2001; Villac & Scheeres 2003; Astakhov & Farrelly 2004; Paskowitz & Scheeres 2006; Granvik et al. 2012; Haapala & Howell 2013; Qi & de Ruiter 2018), we categorize these motions into three types: TCs, low-energy flybys, and collisions. Below, we define each type of motion in detail

(i) TC occurs when an object, such as an asteroid or comet, temporarily becomes a satellite of a planet due to gravitational forces. This state is transient, as the object eventually escapes due to the influence of other forces, like the gravitational pull of a third body. Following Granvik et al. (2012) and Qi & de Ruiter (2018), we consider TC to occur within a disc centred on Jupiter with a radius three times Jupiter's Hill radius, focusing on planar motion in the Sun–Jupiter PCRTBP model. According to Qi & de Ruiter (2018), an object's motion is classified as a TC if it meets the following criteria

(a) The object enters the disc region from outside.
(b) Under the PCRTBP model, the object eventually exits the disc region.
(c) The object's trajectory within the disc has at least two pericentres relative to Jupiter before it exits.

From the perspective of an inertial frame, this implies that the object completes at least one revolution around Jupiter. One revolution is



(a) Local maxima constituting the LCS$_+$.

(b) Fitted surface of LCS$_+$.

(c) Local maxima constituting the LCS$_-$.

(d) Fitted surface of LCS$_-$.

Figure 8. LCS$_+$ and LCS$_-$ with different C.

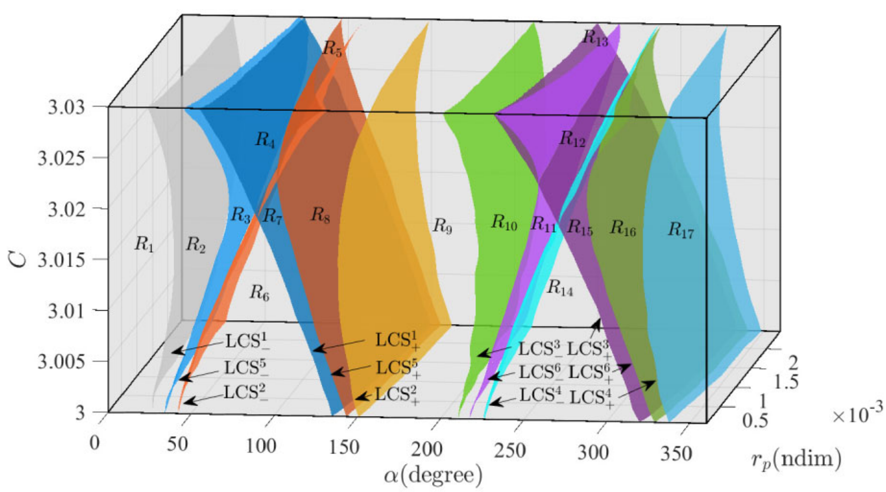
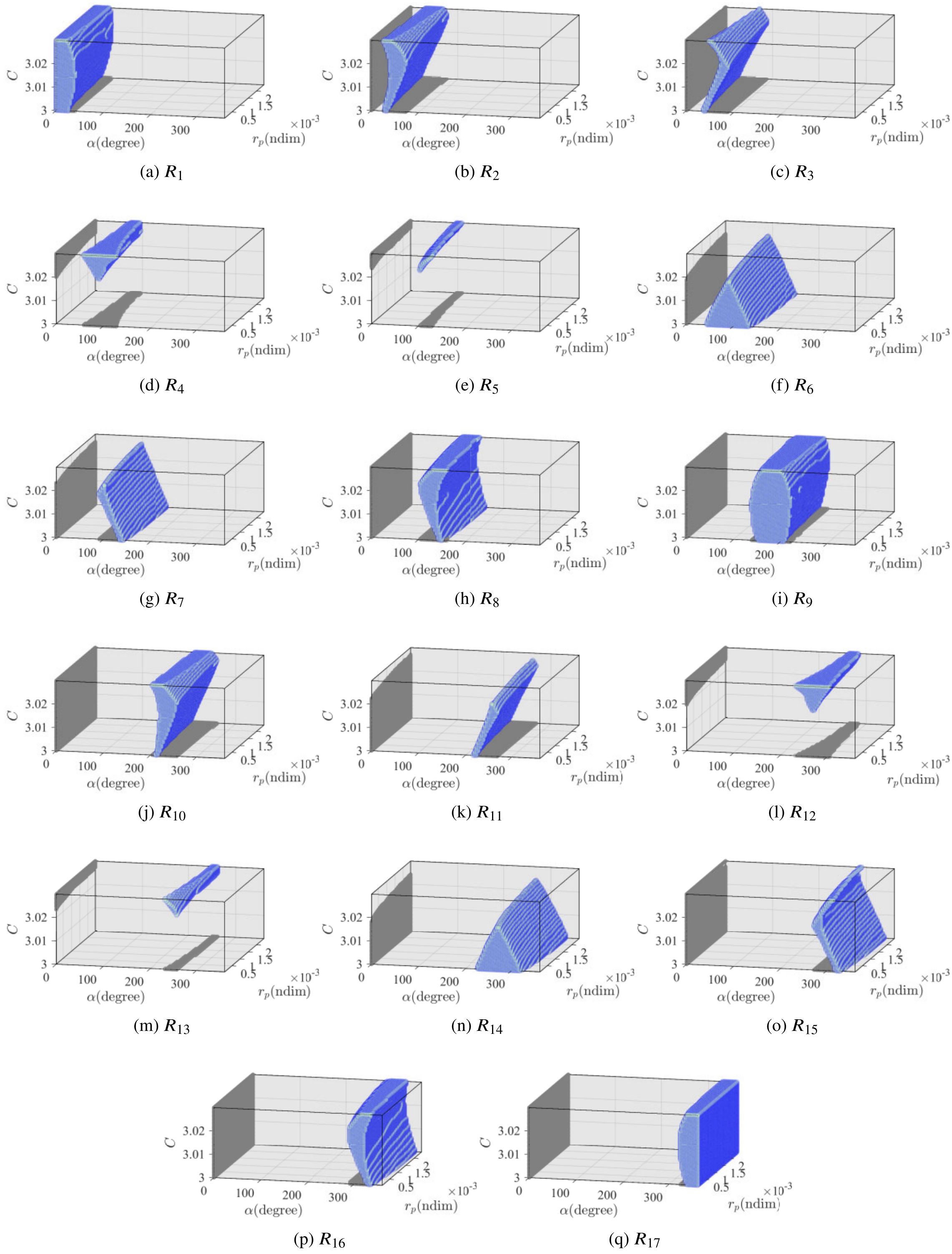


Figure 9. Phase space partitioned based on LCSs.

defined as the segment of its trajectory between two consecutive pericentres.

(ii) The low-energy flyby occurs when the object's trajectory within the disc region includes only a single pericentre relative to Jupiter. Similar to TC, the object enters the planetary system from outside the disc and eventually exits.

(iii) The collision is defined as a scenario where the object's trajectory, when propagated forward or backward from its periapsis state, results in a collision with Jupiter. In our analysis, the normalized integration time $|\Delta T|$ is initially set to 6. If no collision occurs within this period, $|\Delta T|$ is gradually increased until either the trajectory leads to a collision or it reaches the boundary of the disc region.



(a) R_1 (b) R_2 (c) R_3 (d) R_4 (e) R_5 (f) R_6 (g) R_7 (h) R_8 (i) R_9 (j) R_{10} (k) R_{11} (l) R_{12} (m) R_{13} (n) R_{14} (o) R_{15} (p) R_{16} (q) R_{17}

Figure 10. Specific subsets of the phase space partitioned by LCSs.

4.2.2 Investigation of the types of motion in LCS-partitioned subspaces

This subsection investigates the characteristics of object motion corresponding to different regions depicted in Fig. 10. For a point $(\alpha, r_p, C_0)_k$ within region R_s (where $s = 1, 2, \ldots, 17$), the state $\boldsymbol{x}_k(t_0) = [\boldsymbol{r}_k(t_0), \boldsymbol{v}_k(t_0)]$ of the object at its pericentre relative to Jupiter in the Sun–Jupiter BCR frame is determined using equations (11)–(14). The state $\boldsymbol{x}_k(t_0)$ is then propagated forward in the PCRTBP model over a time interval ΔT and backward over $-\Delta T$, where $\Delta T > 0$. Subsequently, we analyse the geometric features of the forward and backward trajectories by selecting various points $(\alpha, r_p, C_0)_k$ within region R_s.

Table 1. Different types of motion for objects near Jupiter.

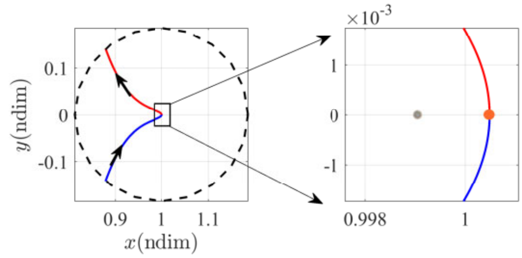
Low-energy flyby	Temporary capture with $np = 2$	Temporary capture with $np > 2$	Collision
R_1, R_6, R_9, R_{14}, R_{17}	R_2, R_8, R_{10}, R_{16}	R_3, R_5, R_7, R_{11}, R_{13}, R_{15}	R_4, R_{12}

Based on whether these trajectories conform to the definitions of TC, low-energy flyby, or collision presented in Section 4.2.1, the corresponding points of these trajectories can be classified into their appropriate types of motion. Consequently, we can ascertain the types of motion represented by the majority of the periapsis states within region R_s. After thorough investigation, the types of motion corresponding to regions R_1 through R_{17} are summarized in Table 1. Here, np denotes the number of pericentres relative to Jupiter for the object's trajectory within the disc region. For instance, in region R_1, the initial parameters are set as $r_p = 1.442 \times 10^{-3}$, $C_0 = 3.000$, and $\alpha = 0°$. The corresponding trajectory in the Sun–Jupiter BCR frame is illustrated in Fig. 11(a). In this figure, the blue trajectory represents the path propagated backward from the periapsis state $\boldsymbol{x}(t_0)$, while the red trajectory indicates the forward propagation path. The black curve outlines the boundary of the disc region, showing that the trajectory within this region has only one pericentre relative to Jupiter, marked by a solid orange circle.

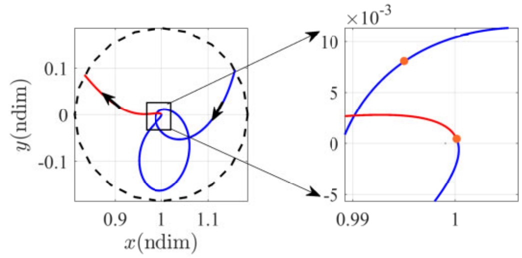
In region R_2, with parameters $r_p = 1.177 \times 10^{-3}$, $C_0 = 3.0021$, and $\alpha = 25.455°$, the corresponding trajectory is shown in Fig. 11(b). This trajectory clearly has two pericentres relative to Jupiter, indicating that in the inertial frame, the object completes one revolution around Jupiter and experiences a weak capture.

Adjusting the parameters to $r_p = 1.730 \times 10^{-3}$, $C_0 = 3.0133$, and $\alpha = 58.182°$ for region R_3 yields the trajectory shown in Fig. 11(c). Although this trajectory also falls under the category of TC, it is noteworthy that the number np of pericentres within the disc region relative to Jupiter exceeds two.

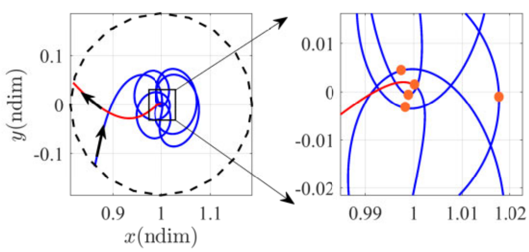
Lastly, Fig. 11(d) illustrates a collision trajectory from region R_4 with parameters $r_p = 9.442 \times 10^{-5}$, $C_0 = 3.0242$, and $\alpha = 87.273°$. As indicated in this figure, regardless of whether the trajectory is propagated backward or forward, it ultimately results in a collision with Jupiter, marked by a solid red circle at the collision point.



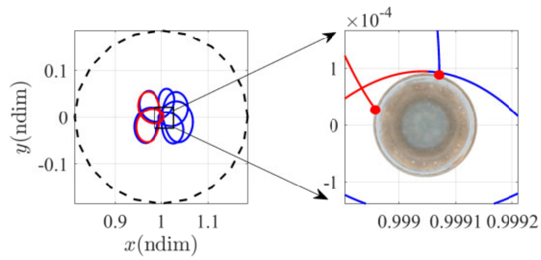
(a) The low-energy flyby trajectory in R_1.



(b) The temporary capture trajectory in R_2.



(c) The temporary capture trajectory in R_3.



(d) The collision trajectory in R_1.

Figure 11. The sample trajectories for objects near Jupiter.

4.3 Analysis of the impact of periapsis parameters and maneuvers on types of motion

4.3.1 The impact of periapsis maneuvers on types of motion

In Section 4.2, we analysed the geometric structures of low-energy flybys, TCs, and collisions in the phase space (α, r_p, C_0). As shown in Fig. 10, the geometric structures corresponding to different types of motion are sensitive to the phase angle α and the Jacobi constant C_0, but not to the periapsis distance r_p. This is evidenced by the significant changes in the projection of the geometric structures on the (C_0, α) plane with variations in α and C_0, while the shape of the projection remains largely unchanged with variations in r_p. For instance, in Figs 10(f) and (n), the projections of regions R_6 and R_{14} on the (C_0, α) plane maintain an approximately triangular shape with only minor changes as r_p varies.

However, C_0 can be adjusted through periapsis maneuvers. An example is illustrated in Fig. 12, where point B_1 initially resides in region R_5, a TC region. By performing an impulsive maneuver at the periapsis relative to Jupiter at time t_0, the Jacobi constant C_0 is altered while keeping α and r_p constant. This maneuver causes point B_1 to move in the phase space (α, r_p, C_0) along the $+C_0$ or $-C_0$ direction.

If the impulsive maneuver increases the object's velocity along the direction of the periapsis velocity vector, C_0 decreases, and point B_1 shifts along the $-C_0$ direction until it intersects with region R_6, which corresponds to low-energy flyby motion. At this intersection, marked as point B_2, the object's motion type transitions from TC to low-energy flyby. Further increases in the maneuver will cause B_2 to continue moving along the $-C_0$ axis, but as shown in Fig. 12, point B_2 will remain within region R_6, meaning the motion type stays as a low-energy flyby.

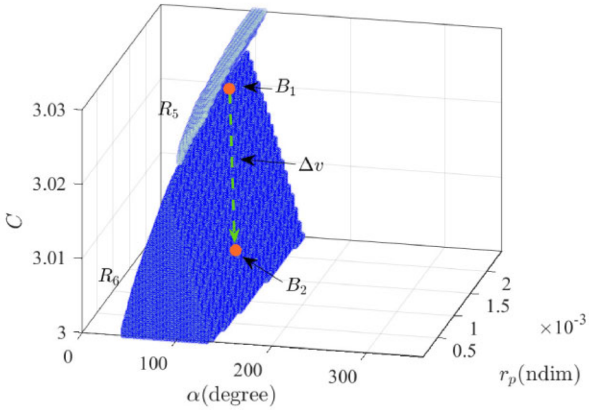


Figure 12. An example of maneuvering to alter the phase space structure.

Conversely, if the maneuver is performed in the opposite direction to the periapsis velocity vector, point B_2 will move along the $+C_0$ direction. When B_2 reaches the location of B_1, the motion type will change from low-energy flyby back to TC. This analysis demonstrates that while the geometric shape of region R_s is significantly influenced by α and C_0, only C_0 can be adjusted via maneuvers, whereas α remains unaffected.

4.3.2 *The impact of the polar angle on types of motion*

To analyse the impact of the polar angle α on the motion of objects near Jupiter, we selected a cross-section at $C_0 = 3.00$. The regions R_s are projected onto this plane and combined according to the types of motion listed in Table 1. The results are presented in Fig. 13.

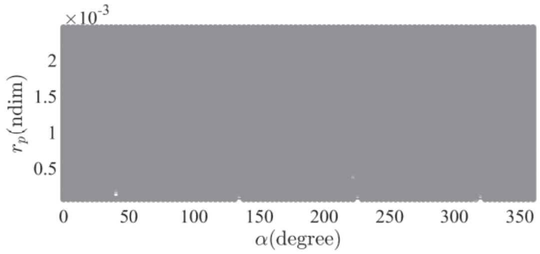
Fig. 13 illustrates the intersections of the projections for regions R_1, R_6, R_9, R_{14}, and R_{17} on the cross-section where the Jacobi constant $C_0 = 3.00$, indicated by the grey area. This grey area occupies a significant portion of the phase space (α, r_p), suggesting that any point within this space could correspond to low-energy flyby motion, provided that periapsis maneuvers are feasible. In Fig. 13(b), the grey area symbolizes TC motion, defined by having two pericentres within the disc region. This area represents the intersection of the projections for R_2, R_8, R_{10}, and R_{16} on the cross-section with $C_0 = 3.00$. Notably, the grey area shows minimal variation along r_p, indicating that the periapsis distance r_p has a negligible effect on TC motions. The unshaded region in Fig. 13(b) signifies that despite any modifications to C_0 through periapsis maneuvers, points within this region cannot transition to TC. We designate this area as the 'prohibited region' for TC, which approximately corresponds to the union of the following subsets: $0^\circ < \alpha < 14^\circ$, $72^\circ < \alpha < 98^\circ$, $152^\circ < \alpha < 200^\circ$, $258^\circ < \alpha < 287^\circ$, and $345^\circ < \alpha < 360^\circ$.

For TC motions involving more than two pericentres, represented by regions R_3, R_5, R_7, R_{11}, R_{13}, and R_{15}, their intersections on the cross-section with the Jacobi constant $C_0 = 3.00$ are shown in Fig. 13(c). Similar to previous findings, we identify a prohibited region for these motions, which aligns with the union of the following subsets: $0^\circ < \alpha < 21^\circ$, $152^\circ < \alpha < 207^\circ$, and $338^\circ < \alpha < 360^\circ$.

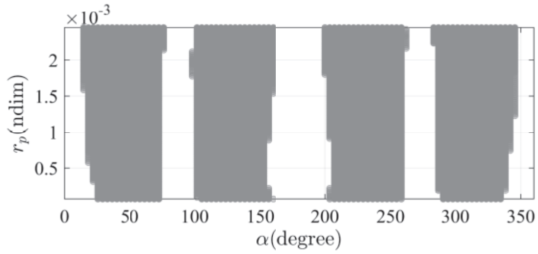
Finally, we examined collision motions for regions R_4 and R_{12} on the cross-section with $C_0 = 3.00$, as illustrated by their intersection in Fig. 13(d). This region is significantly influenced by the periapsis distance r_p, with larger periapsis distances leading to smaller collisional areas. This suggests that a closer approach to Jupiter enhances the probability of collision. Our analysis further indicates that an object's trajectory is most likely to collide with Jupiter within the polar angle ranges $58^\circ < \alpha < 123^\circ$ or $232^\circ < \alpha < 305^\circ$. Additionally, from the subsets representing collision motions, R_4 and R_{12}, depicted in Figs 10(d) and (l), it is evident that for Jacobi constants $C_0 > 3.019$, objects are more likely to collide with Jupiter.

5 POTENTIAL TEMPORARY CAPTURE OBJECTS OF JUPITER

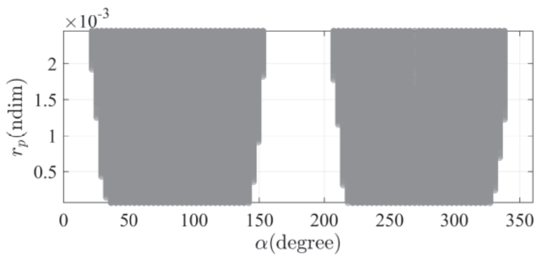
This section introduces the method for identifying objects that can be temporarily captured by Jupiter. This approach is developed based on the TC trajectories in the PCRTBP model.



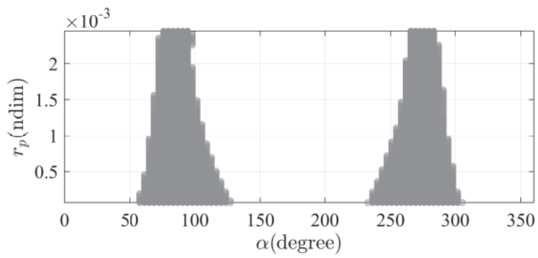
(a) The low-energy flyby region.



(b) The temporary capture region with $np = 2$.



(c) The temporary capture region with $np > 2$.



(d) The collision region.

Figure 13. The projection of the phase space on the section with $C = 3.00$.

Table 2. Orbital elements of the potential TC objects.

Objects	Epoch	a(au)	e	i(deg)	Ω(deg)	ω(deg)	M(deg)
82P/Gehrels 3	2452484.5	4.1354121	0.1232542	1.1265548	239.5914611	227.4956618	38.9936815
P/1996 R2	2450364.5	3.7820399	0.3099908	2.6053420	40.2405612	334.0424860	346.1722312
147P/Kushida–Muramatsu	2454801.5	3.8070858	0.2760252	2.3679636	93.7620655	346.8408342	9.1725771
111P/Helin–Roman–Crockett	2456320.5	4.1603659	0.1096208	4.2288070	89.7939982	3.3779415	359.6946225
74P/Smirnova–Chernykh	2456239.5	4.1655460	0.1477285	6.6510391	77.0842040	86.5771174	139.1496968
39P/Oterma	2456994.5	7.4033251	0.2284123	1.4782370	303.6355887	89.7043305	210.8024450
2002 GV28	2460400.5	3.6197144	0.2968526	1.6303317	258.0657642	352.1501383	45.9321389

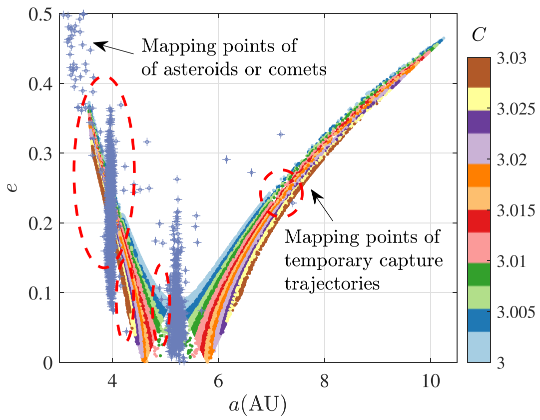


Figure 14. Distribution of a and e of the TC trajectories and the objects.

5.1 An approach for identifying temporarily captured objects by Jupiter via LCSs

This section outlines a method for identifying asteroids or comets that may be temporarily captured by Jupiter. A three-stage screening method for TC objects has been developed. The first stage involves an initial screening of candidate objects based on a coplanar assumption. In the second stage, the geometric characteristics of TC trajectories, obtained from the PCRTBP model, are used to further refine the list of candidates. The final stage subjects the remaining objects to a detailed analysis using an HFE model to confirm their potential for TC by Jupiter.

5.1.1 Preliminary screening of potential objects for temporary capture by Jupiter based on coplanar constraints

In our initial analysis, we assume coplanarity among Jupiter, the Sun, and the object, focusing on the planar motion of the object near Jupiter. Consequently, when examining the TC of comets or asteroids around Jupiter, we restrict the orbital inclination of the objects in the heliocentric J2000.0 ecliptic frame to $0^\circ < i < 5^\circ$, aligning with the coplanarity assumption. Based on the findings of Qi & de Ruiter (2018) regarding short-term capture in the Earth–Moon system, we apply an eccentricity screening criterion of $0 < e < 0.5$ and a semimajor axis screening criterion of 0 au $< a <$ 8 au. These criteria are broader than those in Qi & de Ruiter (2018), allowing for a wider range of potential TC candidates during the initial screening phase.

Using these screening conditions, we identify 2507 asteroids or comets out of a total of 39 249 candidates that meet the criteria. The initial states of these objects are retrieved from the JPL Small-Body Database (https://ssd.jpl.nasa.gov/sbdb.cgi). Their orbital characteristics, specifically semimajor axes and eccentricities, are visually represented in Fig. 14.

5.1.2 Further screening of potential TC objects based on temporary capture trajectories in the Sun–Jupiter system

As previously mentioned, regions R_2, R_8, R_{10}, R_{16}, R_3, R_5, R_7, R_{11}, R_{13}, and R_{15} are recognized as relevant areas for TC within the phase space (α, r_p, C).

We conduct backward integration to propagate the initial periapsis state $\boldsymbol{x}_k(t_0)$ corresponding to the points $(\alpha, r_p, C)_k$ within these regions, under the Sun–Jupiter PCRTBP model. We then identify the intersection points of the object's trajectory with the boundary of the disc area upon entry into the Jupiter system, denoted as $\boldsymbol{x}_k^{\text{in}}(t_0)$. The state $\boldsymbol{x}_k^{\text{in}}(t_0)$ in the Sun–Jupiter BCR frame is subsequently converted into orbital elements within the heliocentric J2000.0 ecliptic frame, as described by Canales et al. (2021b). Fig. 14 shows the distribution of the semimajor axis a and eccentricity e of TC trajectories along the boundary of the disc region, with colours indicating the magnitudes of the Jacobi constant C of the Sun–Jupiter system.

In Fig. 14, the potential for TC by Jupiter is suggested by the overlap of the objects' mapped points (a, e) with those of known TC trajectories in the PCRTBP model. These overlapping regions are highlighted with elliptical dashed lines in Fig. 14, marking areas of interest for further investigation. Among the objects within these designated areas, 489 asteroids or comets are selected for a more in-depth examination.

5.1.3 Final traversal screening of TC objects in the high-fidelity ephemeris model

To analyse these selected objects, we retrieve their initial orbital states in the heliocentric J2000.0 ecliptic frame from the JPL Small-Body Database. Using the ephemeris model outlined in Section 2.2, we propagate these states forward to 2124 January 1 and backward to 1924 January 1. This dual-direction propagation provides a comprehensive view of each object's trajectory over a two-century span. The trajectories, calculated in the heliocentric J2000.0 ecliptic frame, are then transformed into the Sun–Jupiter BCR frame for detailed analysis. The rotation matrices are detailed in Canales et al. (2021b).

By evaluating whether the trajectories of these celestial bodies meet the criteria for TC as defined in Section 4.2, we can determine their potential for TC by Jupiter. Through this analysis, we identify six comets that can be temporarily captured by Jupiter: 82P/Gehrels 3, 147P/Kushida–Muramatsu, 111P/Helin–Roman–Crockett, 74P/Smirnova–Chernykh, 39P/Oterma, and P/1996 R2. Additionally, an outer main-belt asteroid, 2002 GV28, is identified as a potential candidate for TC by Jupiter. Their orbital elements in the heliocentric J2000.0 ecliptic coordinate system are obtained from the JPL Small-Body Database and are detailed in Table 2. The semimajor axes and eccentricities of these comets and asteroids are shown in Fig. 15, and they fall within the (a, e) region identified as corresponding to TC motions in Fig. 14.

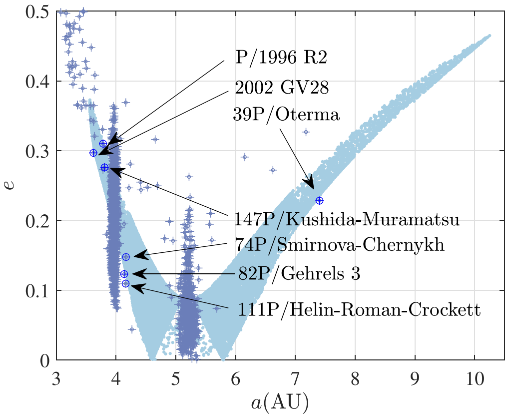


Figure 15. Potential TC objects in (a, e) coordinates.

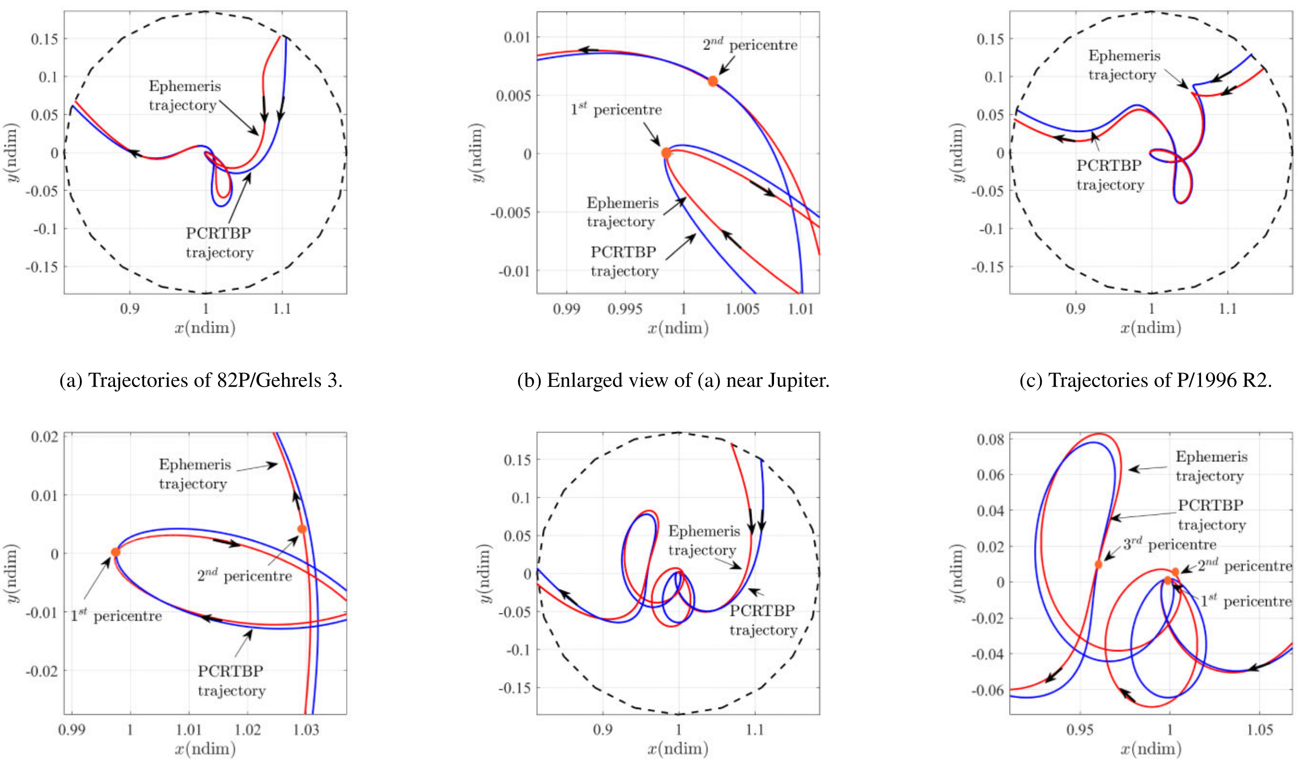
5.2 Analysing trajectories of TC objects

5.2.1 Analysing trajectories of TC comets

Ohtsuka et al. (2008) analysed the TC trajectories of 82P/Gehrels 3, 147P/Kushida–Muramatsu, 111P/Helin–Roman–Crockett, and P/1996 R2 in the Sun–Jupiter rotating frame. Subsequently, Haapala & Howell (2013) identified similar trajectories for 82P/Gehrels 3, 147P/Kushida–Muramatsu, and P/1996 R2 based on the periapsis map of invariant manifolds in the PCRTBP model. However, analysing the trajectory of 111P/Helin–Roman–Crockett poses challenges due to its three-dimensional nature in the Sun–Jupiter rotating frame, making escape and capture dynamics developed under the PCRTBP less applicable. To address this, Haapala & Howell (2013) employed high-dimensional periapsis mapping of invariant manifolds in the spatial CRTBP model to find similar TC trajectories for comet 111P/Helin–Roman–Crockett.

In this paper, we determine the conditions for TC using LCSs, focusing on the distribution of polar angle α, periapsis distance r_p, and Jacobi constant C corresponding to TC trajectories. Under the Sun–Jupiter PCRTBP model, we also identify nearly planar TC trajectories similar to those of comets 82P/Gehrels 3, P/1996 R2, and 147P/Kushida–Muramatsu, as shown in Figs 16(a), (c), and (e), respectively. The paths of these TC comets are obtained by propagating the initial states in Table 2 according to the ephemeris model detailed in Section 2.2, represented by red solid lines. The TC trajectories under the PCRTBP model are shown by blue solid lines, with the boundaries of the disc region indicated by black dashed lines.

It can be observed that the paths of 82P/Gehrels 3 and P/1996 R2 exhibit two pericentres relative to Jupiter. For 82P/Gehrels 3, after entering the Jupiter system, it revolves counterclockwise relative to Jupiter at the second pericentre, indicating direct motion, as illustrated in Fig. 16(b). This motion corresponds to a polar angle $\alpha = 53^\circ$, which falls within the TC region delineated in Fig. 13(b) (where the number of pericentres within the disc region equals two). The trajectory of P/1996 R2 similarly revolves counterclockwise relative to Jupiter at the second pericentre, with a polar angle $\alpha = 14.6^\circ$, also within the grey region in Fig. 13(b) indicating TC. Fig. 16(e) displays the trajectory of comet 147P/Kushida–Muramatsu, which has three pericentres relative to Jupiter within the disc region, meaning it completes at least 1.5 revolutions around Jupiter in the inertial frame.



(a) Trajectories of 82P/Gehrels 3.

(b) Enlarged view of (a) near Jupiter.

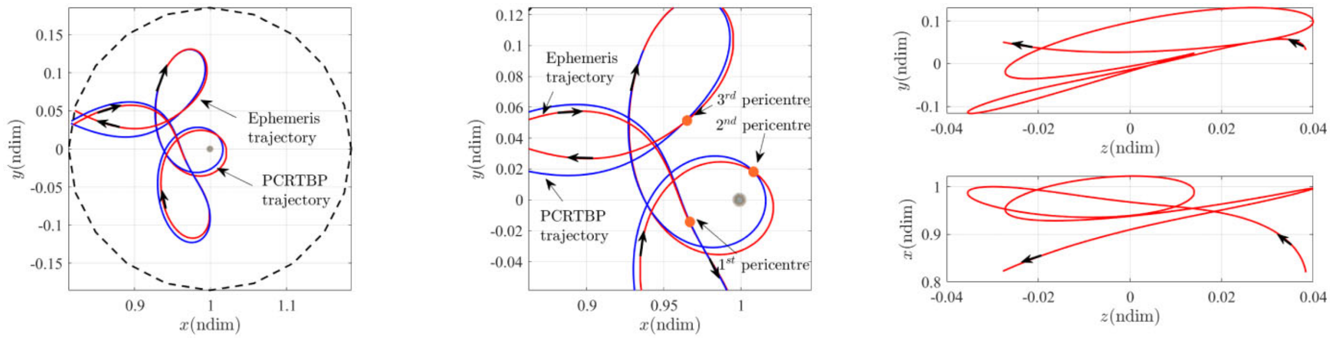
(c) Trajectories of P/1996 R2.

(d) Enlarged view of (c) near Jupiter.

(e) Trajectories of 147P/Kushida-Muramatsu.

(f) Enlarged view of (e) near Jupiter.

Figure 16. Trajectories of the TC objects.



(a) Trajectories of 2002 GV 28.

(b) Enlarged view of (a) near Jupiter.

(c) Projection of the trajectory onto the zy and zx plane.

Figure 17. TC trajectories of 2002 GV 28.

Table 3. 1σ error of the orbital elements for 2002 GV28.

a(au)	e	i(deg)	Ω(deg)	ω(deg)	M(deg)
1.6853×10^{-5}	1.2818×10^{-5}	3.5961×10^{-5}	0.0017011	0.0055618	0.0037544

5.2.2 Analysing trajectories of TC asteroid 2002 GV28

Next, we investigate the TC trajectory of the asteroid 2002 GV28. Fig. 17 illustrates the ephemeris path of 2002 GV28 alongside its counterpart in the PCRTBP model, assuming no initial state uncertainties. The ephemeris path of 2002 GV28 in the heliocentric J2000.0 ecliptic frame is similarly obtained by propagating the corresponding initial state from Table 2 within an HFE model. It is then transformed to display in the Sun–Jupiter rotating frame through a coordinate transformation. The trajectory of 2002 GV28 in the PCRTBP model is derived by propagating its periapsis state within the Sun–Jupiter PCRTBP framework. As shown in Fig. 17(b), the trajectory within the disc region exhibits three pericentres relative to Jupiter, indicating that the asteroid completes at least 1.5 revolutions around Jupiter in the inertial frame. Fig. 17(c) depicts the projections of the TC trajectory on the zx and zy planes, respectively. It is evident that the motion in the z-direction is an order of magnitude smaller than in the x- and y-directions, suggesting that the TC trajectory can be approximated as a planar trajectory.

Additionally, the entry epoch of the asteroid into the disc region, marking the boundary of its capture, is determined to be 2064 April 2 at 13:04:59.13 (UTC). The epoch of escape from the Jupiter system, corresponding to its second arrival at this boundary, is identified as 2078 July 17 at 02:49:25.66 (UTC). This results in a TC duration of 14.2877 yr. Furthermore, the epochs of the asteroid's three pericentres relative to Jupiter are identified as 2065 December 14 at 09:44:49.56 (UTC), 2071 May 20 at 20:36:33.81 (UTC), and 2077 March 24 at 18:16:21.55 (UTC), with respective periapsis distances of 377.60 R_J, 244.35 R_J, and 661.24 R_J, where R_J denotes Jupiter's radius, approximately 71 492 km.

We also investigate the impact of initial state uncertainty on the TC trajectories of 2002 GV28. The 1σ errors of the orbital elements of 2002 GV28 in the heliocentric J2000.0 ecliptic frame, obtained from the JPL Small-Body Database, are shown in Table 3. Subsequently, we selected 200 initial states considering these uncertainties and forward integrated them to obtain 200 TC trajectories, depicted in the disc area in Fig. 18. Evidently, the TC motion of 2002 GV28 is unstable. Compared to the nominal trajectory without initial state deviations (shown in Fig. 17a), some trajectories no longer escape the system towards the Sun, as indicated by the trajectories on the left in Fig. 18. Instead, they escape the Jupiter system away from the Sun, as shown by the trajectories on the right in Fig. 18.

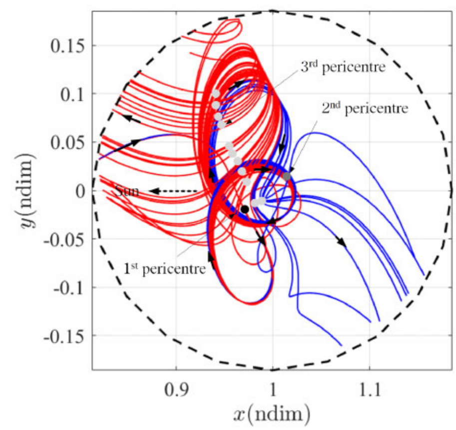


Figure 18. The trajectories of 2002 GV28 considering uncertainties.

Although the perturbed trajectories do not exhibit significant differences from the nominal trajectory at the first and second pericentres, they differ at the third pericentre, where some perturbed trajectories have a third periapsis distance shorter than the first periapsis distance, as indicated by the trajectories shown on the right in Fig. 18. Although the trajectories of 2002 GV28 are significantly affected by initial state uncertainties, all trajectories considering these uncertainties retain three pericentres relative to Jupiter within the disc area, indicating that this asteroid can complete at least 1.5 revolutions around Jupiter in the inertial frame.

6 CONCLUSIONS

In this paper, the TC motion of objects near Jupiter is investigated within the framework of the Sun–Jupiter PCRTBP using LCSs and periapsis Poincaré maps. The 12 LCS surfaces divide the phase space (α, r_p, C) into 17 subsets. Among these, four subsets correspond to TC trajectories with $np = 2$, and six subsets correspond to TC trajectories with $np > 2$. The remaining subsets account for five low-energy flyby motions and two collision motions. TC motion is found to be closely related to the polar angle of the object at pericentre relative to Jupiter. For direct motion, there is a low probability of TC motions occurring for $np = 2$ when $0^\circ < \alpha < 14^\circ$, $72^\circ < \alpha < 98^\circ$, $152^\circ < \alpha < 200^\circ$, $258^\circ < \alpha < 287^\circ$, and $345^\circ < \alpha < 360^\circ$. For $np = 3$, the regions where TC is not possible are $0^\circ < \alpha < 21^\circ$, $152^\circ < \alpha < 207^\circ$, and $338^\circ < \alpha < 360^\circ$. The TC trajectories obtained in the PCRTBP model are projected into the heliocentric inertial frame to determine the distribution of semimajor axes and eccentricities, as represented in the (a, e) diagram. A total of 489 comets and asteroids are identified at the intersection of this diagram with the (a, e) diagram for Jupiter-family objects to be screened. The trajectories of these objects are then analysed using the HFE model. This analysis identifies six known TC comets, including P/1996 R2, 82P/Gehrels 3, 147P/Kushida–Muramatsu, 39P/Oterma, 74P/Smirnova–Chernykh, and 111P/Helin–Roman–Crockett. Furthermore, an outer main-belt asteroid, 2002 GV28, is identified as having the potential to be temporarily captured by Jupiter at 2064 April 2, 13:04:59.13 (UTC). After approximately 14.2877 yr of orbiting Jupiter, this asteroid is expected to escape Jupiter's gravitational influence. Finally, trajectories in the PCRTBP model that resemble the paths of these TC comets and asteroids are identified.

ACKNOWLEDGEMENTS

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DATA AVAILABILITY

The data underlying this article are available in the article and its online supplementary material. The coordinates of the local maxima of FTLEs forming the LCS surfaces in this work are available at https://zenodo.org/records/13712664. One can obtain the initial periapsis states of different types of motion in the Sun–Jupiter PCRTBP model, such as TC, low-energy flyby, or collision, by considering the subsets of phase space partitioned by these LCS surfaces. Propagating these initial periapsis states yields trajectories corresponding to the respective types of motion.

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